

Marketing Funnel & Conversion Performance Analysis Report

Data Science & Business Analytics Internship Project

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Tools Used: Python & Microsoft Power BI

1. Executive Summary

This project analyses customer movement across the marketing funnel to understand how users progress from website traffic to final customer conversion. The analysis focuses on conversion performance, funnel drop-offs, and cart abandonment behaviour to identify opportunities for business improvement.

The business generated a total traffic of **100,000 users** with an **overall conversion rate of 45.98%**. The findings show strong engagement during the lead generation and cart stages, while a significant drop-off occurs during the final purchase stage.

An interactive Power BI dashboard was developed to visualise funnel performance, customer behaviour, and conversion trends.

2. Business Objective

The objective of this project is to evaluate marketing funnel performance and identify factors affecting customer conversions.

The analysis aims to:

- Track customer movement through funnel stages
 - Measure conversion performance
 - Identify major customer drop-off points
 - Analyse cart abandonment behaviour
 - Recommend strategies to improve conversions and revenue
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3. Dataset Overview

The analysis was performed using a marketing funnel dataset containing:

- Website traffic sessions
- Product views

- Cart additions
- Customer conversions
- Bounce status
- Cart value
- Discount percentage

Key Metrics

Metric	Value
Total Traffic	100,000
Traffic to Lead Rate	96.08%
Lead to Cart Rate	96.22%
Cart to Customer Rate	49.73%
Overall Conversion Rate	45.98%
Cart Abandonment Rate	50.27%

4. Data Preparation & Methodology

The dataset was cleaned and structured using Python.

Steps Performed

- Verified data types and missing values
- Created funnel stage indicators:
 - Traffic
 - Lead
 - Cart Lead
 - Customer
- Calculated conversion metrics and drop-off rates
- Prepared dataset for Power BI visualization

Funnel Structure

Traffic → Lead → Cart Lead → Customer

5. Tools Used

Python

Used for:

- Data cleaning
- Funnel calculations
- Conversion analysis
- Drop-off analysis

Microsoft Power BI

Used for:

- Dashboard development
 - KPI visualization
 - Funnel visualization
 - Business reporting
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6. Dashboard Overview

- KPI Summary Cards
 - User Conversion Funnel
 - Funnel Drop-Off Analysis
 - Cart Abandonment Analysis
 - Interactive Slicers
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7. Key Insights

- The highest drop-off occurs during the **Cart Lead → Customer** stage with a drop-off rate of **50.27%**.
- The funnel performs strongly during the **Traffic → Lead** and **Lead → Cart** stages with conversion rates above **96%**.

- Users actively add products to the cart, indicating strong purchase intent and customer engagement.
 - The high cart abandonment rate suggests friction during checkout or payment completion.
 - The checkout stage is the primary area requiring optimization to improve overall conversions.
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8. Business Recommendations

- Simplify the checkout process to improve purchase completion rates.
 - Implement cart reminder emails and promotional offers to reduce cart abandonment.
 - Provide multiple secure payment options for better customer convenience.
 - Improve customer trust through reviews, secure payment badges, and transparent policies.
 - Continuously monitor funnel performance to identify conversion improvement opportunities.
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9. Business Impact

- Optimising the checkout stage can significantly improve customer conversion rates.
 - Reducing cart abandonment can increase overall business revenue and profitability.
 - Improved user experience can enhance customer satisfaction and engagement.
 - Better conversion strategies can strengthen marketing ROI and business growth.
 - Continuous funnel monitoring supports data-driven business decision-making.
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10. Conclusion

The analysis shows strong performance during the lead generation and cart engagement stages. However, significant customer loss occurs during the final purchase stage due to high cart abandonment.

By optimising the checkout experience and reducing funnel drop-offs, the business can improve conversion efficiency, customer acquisition, and overall revenue growth.